Grounded Workflow Composition for Fast and Reliable In-Car Agents

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Abstract

CAR-bench evaluates whether tool-using in-car assistants complete multi-turn requests while respecting policy, ambiguity, and missing-capability constraints. We present a Track 2 agent that uses Cerebras-hosted gpt-oss-120b [OpenAI, 2025; Cerebras Systems, 2026] for bounded semantic interpretation and workflow composition, while delegating fact management, policy checks, calculations, and side effects to deterministic components. The agent converts each substantive user turn into typed action obligations, assigns each obligation to a compatible workflow, executes only ready obligations, and accepts completion only when grounded tool evidence matches the requested target. Declarative action, policy, fact, and workflow graphs reduce prompt size and prevent brittle planner competition. A shared sequential-call budget enforces the five-call Track 2 constraint. The frozen submission passes 1,918 unit tests plus 140 generated subtests. A one-trial campaign over all 254 public scenarios passed 145 (57.1%) in the raw result; 16 scenarios ended with provider or official-simulator errors before usable Agent evidence. We report the raw score and its evidence boundary rather than hidden-set performance.

1 Introduction

CAR-bench targets a difficult reliability boundary: an assistant must interpret short, context-dependent requests, call interconnected tools, obey domain policies, and decline requests when the environment lacks the required capability or fact [Kirmayr et al., 2026]. Hallucination and disambiguation tasks make a monolithic "prompt plus all tools" agent particularly fragile. Route and POI payloads consume context, while a small mistake in confirmation, preference priority, target identity, or tool ordering can invalidate an otherwise plausible answer.

Our design separates probabilistic language understanding from deterministic execution. The language model resolves conversational meaning and, only when necessary, chooses among a bounded set of workflow compositions. Engineering components remain authoritative for observed facts, tool capabilities, policy predicates, numeric calculations, workflow ownership, and completion evidence. This division is intended to retain linguistic generality without allowing an LLM to invent identifiers, silently change goals, or freely improvise tool chains.

The main contributions are:

- a context-aware semantic framing layer that produces per-action typed intermediate representations (IRs) and readable canonical requests;
- an obligation ledger and constrained workflow composer that preserve action ownership and dependencies across dialogue turns;
- scoped fact and policy graphs with deterministic executors and evidence-bound completion contracts; and
- selective context construction, tool-result compaction, and a shared sequential-call budget for fast Track 2 inference.

2 Architecture

Figure 1 summarizes one baseline LLM step. An incoming A2A message received through A2A [A2A Protocol Project, 2026] first passes through a narrow protocol parser. Only replies bound to existing state, such as confirmation, an ordinal selection, or a requested numeric value, are handled deterministically. Every other substantive request requires semantic framing; framing failure cannot fall through to side-effect execution based on raw text.

2.1 Semantic Framing and Typed Actions

The framing prompt receives the current request, a bounded natural-language transcript, active goals, relevant scoped facts, currently referable candidate sets, and a small set of retrieved policy hints. It does not receive every tool schema or historical tool payload. The model emits a canonical natural-language request and a

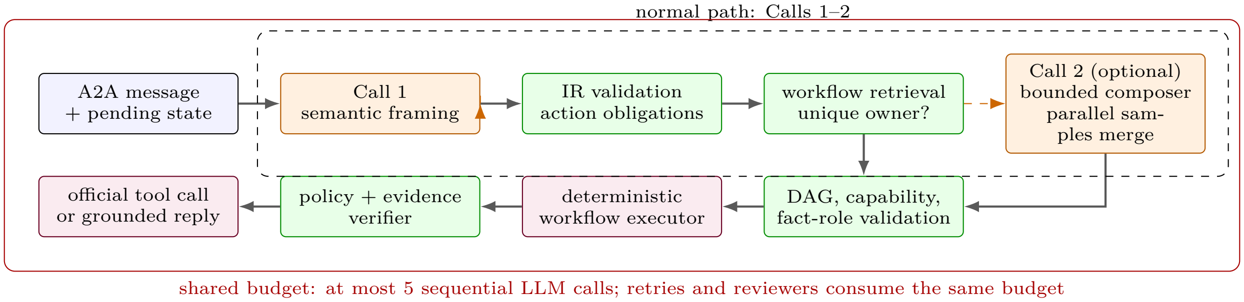

Figure 1: Track 2 architecture and call accounting. A unique owner bypasses Call 2; ambiguous compositions may parallel-sample and merge without increasing depth. The small dashed loop shows malformed-output repair and revalidation. Narrow ownership/final-output recovery uses Calls 4–5 at most; all branches and retries share the same five-call budget.

list of semantic actions. Each action includes a canonical intent, operation, target type, evidence spans, dependencies, and typed slots. Canonical enums constrain the output vocabulary.

The engineering layer assigns stable action IDs and validates fields independently. Invalid fact references or unsupported enum values are removed without discarding a valid intent. Complex or conflicting slots may trigger one narrow batch slot-resolution call after workflow contracts are known. This prevents an ordinal such as "the second one" from being attached to the wrong action type. Literal text and regular expressions are restricted to bounded protocol replies and validated compatibility adapters; they cannot authorize an open-ended intent or side effect.

2.2 Obligations, Ownership, and Composition

Validated actions compile into persistent obligations. Each obligation records its target fact roles, slots, dependencies, constraints, owner workflow, status, and completion contract. Owner assignment is deterministic when exactly one workflow contract is compatible. Multi-action, cross-domain, conditional, or multiple-owner requests invoke a narrow composer that selects and orders one to four existing workflows rather than generating arbitrary tools.

The composition validator checks action coverage, accepted fact roles, input availability, produced facts, side effects, and branch predicates. Ownership is sticky across turns: a reply that fills a missing contact or selection updates the same obligation instead of restarting global workflow selection. The scheduler advances ready obligations whose dependencies are satisfied. The contract records typed failures per action; however, Section 6 identifies a frozen-scheduler edge case in preserving independent siblings after such a failure.

3 Grounded Execution

3.1 Fact and Capability Boundaries

Official tool results are parsed once into a scoped fact store. A fact records its stable ID, source tool and arguments, session/workflow scope, semantic role, and selected-set relationships. ResolvedTarget provides a uniform handoff for POIs, routes, contacts, locations, and calendar events. For example, a user's "second restaurant" selection becomes a specific POI fact with role navigation_destination; downstream navigation consumes its ID rather than re-querying the restaurant name as a city.

Large route and POI results are compacted to decision-relevant fields. Old raw payloads are omitted from model prompts once represented by facts, while a wider bounded history of user and assistant natural-language messages is kept to preserve conversational intent. Tool routing exposes only the schemas needed by the selected workflow.

Capability checks use the actual tool inventory and observed result fields. Missing tools and missing fields produce typed failures, not guessed values. Dependent obligations stop or request clarification. This prevents fabricated results when a Hallucination scenario removes a supported capability, although independent-sibling continuation is incomplete in the frozen runtime.

3.2 Policy and Deterministic Workflows

The policy graph is the authority for represented rules, including trigger conditions, preference priority, required observations, parameter constraints, repairs, refusals, and response obligations. Python implements generic predicates and graph execution rather than dispatching on task IDs or city names. Before a configurable action, the agent retrieves relevant preferences and applies the official priority ordering: explicit user instruction, personal preference, heuristic default, current context, then clarification.

The workflow library contains 37 workflow definitions built from 57 atomic capabilities, with 26 declarative workflow state graphs. Deterministic executors handle policy-sensitive sequences and arithmetic, including defrost interlocks, climate configuration, navigation mutations, route/POI handoffs, SOC range and charging-time calculations, and calendar/contact/email chains. The LLM determines meaning and composition but does not calculate an SOC-derived search radius or manufacture a tool ID.

3.3 Verification and Completion

Candidate actions pass through a post-selection verifier. It checks that the tool belongs to the owner workflow, its arguments bind to the obligation's facts, required policy observations exist, and no unrequested side effect is introduced. Ambiguous ownership may use one narrow reviewer supplied with the conflicting candidates and a short relevant transcript. The verifier does not reinterpret the original request or choose a new workflow.

Completion is evidence-bound. A successful tool with the right name does not close an arbitrary obligation: tool arguments, target facts, result fields, and typed predicates must satisfy its completion contract. A grounded response composer reports only facts supported by official results and required policy language. Pending obligations may cause an intermediate clarification or option presentation, but the agent cannot claim the whole task is complete.

4 Illustrative Execution

Consider a multi-turn request to find restaurants in Barcelona, select the second result, and replace an active navigation destination with that POI. The first turn produces a poi.search obligation. Its deterministic executor resolves Barcelona through the official location tool and stores the returned restaurants as a scoped candidate set. The Agent presents concise options without retaining the full POI payload in later prompts.

When the user says "the second one," the protocol layer can resolve the ordinal only because the workflow is waiting for a POI selection and exactly one current POI candidate set is in scope. The resulting fact contains the selected POI ID, its source result, candidate index, and intended role. A dependent navigation.edit obligation becomes ready. Its owner consumes the POI ID directly, retrieves compatible routes when required, and invokes the official navigation mutation. Completion requires evidence that the mutation's destination argument equals the selected POI fact. The system cannot silently reuse an older Paris search, reinterpret the restaurant as a city, or close the request after merely presenting a route.

This example exercises the general producer-consumer contract used elsewhere: calendar events can produce locations or attendee sets; contacts can produce email recipients; routes can produce constrained charging-search scopes; and vehicle-state probes can satisfy policy prerequisites. Each handoff is typed, scoped, and checked before a side effect.

4.1 Failure and Recovery Semantics

Recovery is represented as data rather than inferred from error prose. A failure distinguishes transient transport, malformed model output, invalid fact reference, ambiguous semantics, missing capability, missing result field, policy prohibition, and exhausted sequential budget. Recovery then has a bounded disposition: retry transport without claiming a new reasoning result, repair only the invalid IR fields, query an available read-only prerequisite, ask the user when multiple meanings remain, or return a grounded inability statement. Independent work continues only when the frozen scheduler preserves the sibling path.

The no-progress guard tracks repeated action signatures and typed failures. It stops loops in which a fallback repeatedly queries the same state or proposes the same rejected tool call. It does not suppress legitimate repeated tools when their target facts differ, as in two route legs or multiple requested vehicle zones.

4.2 Policy-Sensitive Workflow Examples

Vehicle controls illustrate why the policy graph and executor are separate from semantic framing. A defrost request is a single semantic action, but its workflow can require climate and window observations, closing windows beyond a policy threshold, enabling air conditioning, raising fan speed, and finally activating the correct defrost target. The framer need not reproduce this sequence. It identifies the requested effect and scope; graph predicates derive the required stages from current official state. A sunroof request similarly keeps sunroof position, sunshade position, weather risk, requested percentage, and confirmation state as distinct facts so that one target cannot overwrite another.

Navigation and charging workflows use the same principle for numeric and route constraints. An SOC buffer is typed separately from a charging target SOC. Official calculator results bind the reachable distance and estimated charging time. POI searches along a route must consume that observed distance and the active route fact, while route selection must honor an explicit road or route constraint. If no candidate satisfies a requested constraint, the workflow presents grounded alternatives or asks the user rather than selecting an unrelated default.

Productivity workflows demonstrate multi-action continuity. Calendar results produce event and attendee facts; contact lookup resolves recipients; weather or navigation actions consume only the event location; and email composition consumes the requested content plus recipient facts. Sending an email cannot authorize navigation merely because a POI appears in its body. If contact or email capability is removed, the corresponding obligation is blocked rather than replaced with an invented action. These behaviors follow typed contracts shared across tasks rather than phrases or public scenario identifiers.

5 Fast-Reasoning Strategy

All model-facing subsystems share one SequentialCallBudget. The normal path uses one framing call and zero or one composition call. Conditional calls include malformed-output recovery, narrow IR repair, ownership review, and a final constrained fallback. Each consumes the same budget; a sixth sequential call is rejected locally. Network retries that produce no model response

Evidence	Result	Scope
Local tests	1,918 + 140 pass	frozen source
Base	48/100	raw, one trial
Hallucination	77/98	raw, one trial
Disambiguation	20/56	raw, one trial
Public campaign	145/254	raw, one trial
Valid Agent evidence	145/238	classified
Agent tokens	11,349,790	44,684/task
A2A turn latency	2.77 s	1,313 turns
Digest pull + A2A card	pass	release image

Table 1: Frozen evidence. Sixteen infrastructure failures remain in the raw denominator; results are neither Pass^3 nor hidden-set scores.

are tracked separately from successful reasoning calls. A2A turn metrics aggregate prompt, completion, and thinking tokens across every model invocation.

Speed comes from avoiding repeated large prompts rather than minimizing model use at any cost. The agent selects domain tools, retrieves a few policy nodes, projects runtime markers, compacts tool results, and injects only facts related to the current obligations. Deterministic calculations and workflow stages avoid additional calls for steps whose outcome is fixed by official data. Model name, reasoning effort, token cap, service tier, timeout, temperature, and retry parameters remain configurable through environment variables. Reported public measurements used gpt-oss-120b at medium reasoning effort through the competition Cerebras organization. Latency is observed A2A Agent time, not end-to-end evaluator time or a hardware-independent throughput claim.

6 Validation and Lessons

We used public tasks as contract probes, not sources of answer-specific rules. Recurring failures involved fact roles, owner continuity, premature completion, missing-capability scheduling, and oversized route/POI history. Explicit layer boundaries and compacting tool payloads were more effective than a larger prompt; retaining a bounded natural-language transcript preserved pronouns and task continuation.

The architecture maps directly to CAR-bench's six subscores. Completion contracts target r_actions_final; the side-effect verifier targets r_actions_intermediate; policy prerequisites target r_tool_subset; typed schemas and executors target r_tool_execution_errors; the policy graph and verifier target r_policy_errors; and capability/result guards plus grounded replies target r_user_end_conversation.

6.1 Ablations and Design Lessons

Our development history provides qualitative ablations. Exposing all 57 tool schemas and replaying raw route/POI payloads increased prompt size and made policy details easier to miss; domain-scoped tools plus fact compaction reduced both effects. Allowing specialized planners to compete in a long priority chain caused unrelated workflows to seize ownership; contract-first retrieval and a single selector removed this order dependence. Re-running intent parsing after every short follow-up caused owner drift; persistent obligations converted such turns into slot or fact updates. Finally, tool-name-only completion incorrectly ended multi-action requests; target-bound evidence and an obligation ledger kept unfinished sibling actions visible.

The LLM remains important. Pure lexical routing failed on paraphrases, negation, pronouns, and multiple actions, while deterministic parsing of raw open language sometimes overrode a correct model interpretation. The final boundary uses LLM framing for open language and reserves deterministic parsing for state-bound protocol replies. Conversely, using the LLM to calculate ranges or infer missing result fields was less reliable than official tools and typed guards. The design therefore increases model responsibility for semantics but decreases its authority over facts and side effects.

Table 1 covers frozen revision 5746b7dc. One trial on all 254 public scenarios passed 145 (57.1%) raw: 48/100 Base, 77/98 Hallucination, and 20/56 Disambiguation. Sixteen zero-reward runs ended with provider or official-simulator errors before usable Agent evidence and remain in the denominator; excluding only them yields 145/238 (60.9%). Agent usage averaged 44.7K tokens/task, below the 500K constraint. Across 1,313 timed A2A turns, mean latency was 2.77 seconds, excluding evaluator deliberation. We report neither Pass^3 nor a hidden score. Our coverage audit counts the 57 environment tools exposed by the starter's runtime ALL_TOOLS; the competition website's benchmark-level count is 58, including the imported but runtime-disabled fuel-information capability.

6.2 Limitations

The declarative catalogs are extensive but remain manually curated against the official tool and policy contracts. A novel hidden combination can still expose an incomplete workflow input/output contract or an untested branch. Some compatibility adapters remain at the composition root, although static audits prevent them from authorizing open-ended requests from raw text. Public evidence shows that disambiguation and long multi-action continuity remain substantially weaker than missing-capability handling. In the frozen implementation, one missing capability can still terminate a compound request before an independent sibling action runs, limiting grounded partial completion in compound requests. These risks motivate contract-generated simulation and post-competition evaluation rather than stronger performance claims.

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